Ritesh Somashekar

Software Engineer | AI Products

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PROFESSIONAL SUMMARY

- **Software Engineer** with **5+ years** shipping **data, ML, and LLM-powered products** end to end, from research prototype to production, across healthcare, public safety, and research; published at **ACM WSDM '26** and **IEEE PST 2026**.
- Builds **AI applications** with **LLM APIs (GPT-4, Claude, Gemini, Azure OpenAI)**, **prompt engineering**, **structured outputs**, and **RAG**, and tests them for **hallucinations, safety risks, and failure modes** before users see them.
- Partners with researchers, public agencies, and product teams to turn open questions into tools people use, including **civic-tech** AI training software built with **Fairfax County 9-1-1**.

TECHNICAL SKILLS

Languages: Python, SQL, Java, R

Web & Backend: FastAPI, REST APIs, Streamlit, SQLite, PostgreSQL (pgvector), Docker, Kubernetes, CI/CD, Git

AI & LLMs: LLM APIs (GPT-4, Claude, Gemini, Azure OpenAI), Prompt Engineering, Structured Outputs, RAG, LangChain, LlamaIndex, Vector Databases (ChromaDB, Pinecone, FAISS), Llama 3, Ollama, Agentic AI

AI Evaluation & Safety: LLM Evaluation, RAGAS, Hallucination Detection, Guardrails, Ablation Studies, Failure-Mode Analysis, A/B Testing

Data & Research Tools: Pandas, Apache Airflow, Spark/PySpark, Snowflake, Databricks, MLflow, Tableau, Power BI

Machine Learning & Cloud: PyTorch, Scikit-learn, Hugging Face, BERT, XGBoost, AWS (S3, Lambda, Glue, SageMaker, SQS), Azure

EDUCATION

Master's in Data Analytics and Engineering | George Mason University, Fairfax, VA, USA | GPA: 3.93/4.0

Bachelor's in Information Science and Engineering | Visvesvaraya Technological University, Bengaluru, India

PROFESSIONAL EXPERIENCE

George Mason University | ***AI & ML Research Associate*** **Fairfax, VA | August 2024 - Present**

- To scale 9-1-1 call-taker training beyond instructor role-play, co-developed **SHIELD** with **Fairfax County DPSC**, a web app where **LLM agents** simulate calls and coach trainees live; published at **ACM WSDM '26**.
- To move SHIELD from prototype to pilot, built its **Python/Streamlit** interface with **SQLite** accounts, course enrollment, and session tracking, iterating on feedback from instructors and agency testing sessions.
- To generate safe training calls without commercial APIs, co-developed **MATE-911**, a **RAG** application on **Llama 3.1 8B** reaching **0.81 protocol F1** and a **3.0% hallucination rate**; published at **IEEE PST 2026**.
- To give instructors trustworthy content, led the **data pipeline** turning unstructured call records into **160 gold-standard scenarios**, cutting hallucinations from **13.0% to 3.0%**.
- To stop unsafe outputs before they reached trainees, led the **validation layer** with **protocol guardrails** and **hallucination checks**, lowering hallucinations from **9.5% to 3.0%**.
- To learn what actually drove quality, ran **ablation studies** that exposed failure modes, showing **semantic search** retrieval lifted the pass rate from **49.7% to 97.6%**.
- To keep live sessions responsive, engineered **prompt and context assembly** for **Llama 3.1 8B**, delivering responses in under **1 second** for text and **1.3 seconds** for voice.
- To speed up literature review, built a **RAG research assistant** with **LlamaIndex**, **pgvector**, and a **FastAPI** backend, raising **recall@10 from 61% to 88%** for **40+ researchers** at **190 ms p95** latency.
- To compare LLM providers, integrated **GPT-4, Claude,** and **Gemini** APIs through **LangChain** with **structured outputs**, raising answer accuracy by **16%** and cutting hallucinations by **19%**.
- To ship research prototypes reliably, set up **MLflow**, **Docker**, and **CI/CD**, cutting release cycles from **2 weeks to 5 days** with **A/B testing** of model and prompt versions.
- To support disaster-response research, built **Python** and **R** analysis workflows on the **ORC Hopper HPC cluster**, forecasting community coping trends with an **RMSE of 0.89**.

Carelon Global Solutions (Elevance Health) | ***Machine Learning Engineer*** **Bengaluru, India | May 2022 - August 2023**

- To speed up member support, built a **Generative AI** assistant with **Azure OpenAI**, **LangChain**, and **Pinecone** for **15K+ daily queries**, cutting resolution time by **38%** and raising ticket deflection by **25%**.
- To automate clinical text review, built **BERT/BioBERT** extraction models on unstructured medical notes, reaching **80% accuracy** and cutting processing time by **55%**.
- To serve models inside clinical applications, deployed services with **Docker, Kubernetes,** and **Jenkins CI/CD** behind **REST APIs**, improving uptime by **20%** and enabling real-time EHR integration.
- To keep production models accurate over time, set up **monitoring**, **drift detection**, and automated retraining, sustaining **over 95% accuracy** for **12 months** across **200+ end users**.
- To speed up model development, built ingestion of **400K+ XML/JSON healthcare records** into **Snowflake** and **Databricks**, accelerating training by **60%** and cutting preprocessing time by **70%**.
- To feed products reliable data, designed **20+ pipelines** with **Airflow, PySpark,** and **AWS (S3, Lambda, Glue)**, raising throughput by **70%** and reaching **99.9% reliability** under **HIPAA**.

Carelon Global Solutions (Elevance Health) | ***Data Engineer*** **Bengaluru, India | October 2020 - April 2022**

- To reduce manual claims reconciliation, redesigned **claims-balancing workflows** in **Python** and **SQL**, cutting operational overhead by **60%**.
- To eliminate costly invoice errors, automated **medical invoice processing** with rule-based classification, reducing manual errors by **70%**.
- To give business teams faster answers, delivered self-service **Tableau** and **Power BI** reporting tools, cutting reporting latency by **40%**.

PROJECTS

Agentic AI Ticket Automation System | SecureGPT, LLaMA, AWS SQS, JIRA

- To cut security-ticket triage time, built an **agentic AI** system with **SecureGPT** and **LLaMA** on **AWS SQS** that classifies threats and auto-creates **JIRA** tickets, reaching **93% accuracy** and **70%** faster triage.

PUBLICATIONS

- R. S. Nayaka, **R. Somashekar**, K. B. Laskey, L. Wells II, H. Purohit. "SHIELD: A Framework for Online Conversational AI System for Self-Regulated Learning at Emergency Communications Centers." *ACM WSDM '26*. doi:10.1145/3773966.3779411
- R. S. Nayaka, **R. Somashekar**, H. Purohit. "MATE-911: Multimodal AI-powered Training Environment for Next-Generation 911 Dispatcher Preparedness." *IEEE PST 2026*.